

Chapter 7. Factorial Designs and Analysis

7.1 Two-Way Factorial Designs

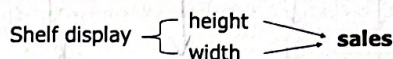
7.2 Multi-Way Factorial Designs

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7.1 Two-Way Factorial Designs

Case Study (1)

- The Castle Bakery Company supplies sliced wrapped Italian bread to a large number of supermarkets in a metropolitan area.
- An experimental study was made of the effects of height of the shelf display (bottom, middle, top) and the width of the shelf display (regular, wide) on sales of this bakery's bread (measured in cases) during the experimental period.



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- Experimental units: Twelve supermarkets, similar in terms of sales volumes and clientele

□ Factors:

- Height of the shelf display with three levels (bottom, middle, top)
- Width of the shelf display with two levels (regular, wide)
- Six treatments (3 × 2)

- Two stores were assigned at random to each of the six treatments, and the display of the bread in each store followed the treatment specifications for that store

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ANOVA Model for Two-Way Factorial Designs

Factor B	B ₁	B ₂	...	B _j	...	B _b
Factor A						
A ₁	AB ₁₁	AB ₁₂	...	AB _{1j}	...	AB _{1b}
A ₂	AB ₂₁	AB ₂₂	...	AB _{2j}	...	AB _{2b}
...	...	...	...	...	...	...
A _i	AB _{i1}	AB _{i2}	...	AB _{ij}	...	AB _{ib}
...	...	...	...	...	...	...
A _a	AB _{a1}	AB _{a2}	...	AB _{aj}	...	AB _{ab}

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk}$$

$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r$

$$SSTO = SSA + SSB + SSAB + SSE$$

$$df_T = df_A + df_B + df_{AB} + df_e$$

这里也可以加入Bk作区及反应会影响自由度。

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Factor A (display height) i	Factor B (display width) j		Row Total	Display Height Average
	B ₁ (regular)	B ₂ (wide)		
A ₁ (Bottom)	47 (Y ₁₁₁) 43 (Y ₁₁₂)	46 (Y ₁₂₁) 40 (Y ₁₂₂)	176 (Y _{1.})	44 (Y _{1.})
Total Average	90 (Y _{11.}) 45 (Y _{11.})	86 (Y _{12.}) 43 (Y _{12.})		
A ₂ (Middle)	62 (Y ₂₁₁) 68 (Y ₂₁₂)	67 (Y ₂₂₁) 71 (Y ₂₂₂)	268 (Y _{2.})	67 (Y _{2.})
Total Average	130 (Y _{21.}) 65 (Y _{21.})	138 (Y _{22.}) 69 (Y _{22.})		
A ₃ (Top)	41 (Y ₃₁₁) 39 (Y ₃₁₂)	42 (Y ₃₂₁) 46 (Y ₃₂₂)	168 (Y _{3.})	42 (Y _{3.})
Total Average	80 (Y _{31.}) 40 (Y _{31.})	88 (Y _{32.}) 44 (Y _{32.})		
Column total Display width average	300 (Y _{.1}) 90 (Y _{.1})	312 (Y _{.2}) 52 (Y _{.2})	612 (Y _{..})	51 (Y _{..})

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Partition of the total sum of square and the degree of freedom

$$\sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{...})^2 = \sum_i \sum_j \sum_k (\bar{Y}_{i..} - \bar{Y}_{...})^2 + \sum_i \sum_j \sum_k (\bar{Y}_{.j.} - \bar{Y}_{...})^2 + \sum_i \sum_j \sum_k (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2 + \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

$abr-1$
 $=$
 $(a-1)$
 $+$
 $(b-1)$
 $+$
 $(ab-a)$
 $(b+1)$

$$SSTO = SSA + SSB + SSAB + SSE$$

$$df_{TO} = df_A + df_B + df_{AB} + df_e$$

$$abr-1 = (a-1) + (b-1) + (a-1)(b-1) + ab(r-1)$$

$\begin{matrix} i & j & k \\ \downarrow & \downarrow & \downarrow \\ i.. & j.. & ij.. \end{matrix}$

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Partition of the total sum of square and the degree of freedom

$$\sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{...})^2 = br \sum_i (\bar{Y}_{i..} - \bar{Y}_{...})^2 + ar \sum_j (\bar{Y}_{.j.} - \bar{Y}_{...})^2 + r \sum_i \sum_j (\bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...})^2 + \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

$$SSTO = SSA + SSB + SSAB + SSE$$

$$df_{TO} = df_A + df_B + df_{AB} + df_E$$

$$abr - 1 = (a-1) + (b-1) + (a-1)(b-1) + ab(r-1)$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
i *j* *ij* *k*

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自由度 = 该效应水平

ANOVA table for two-way factorial designs

Source	Degree of freedom	E(MS) Fixed	E(MS) Random	E(MS) Mixed
A	a - 1	$\sigma_e^2 + \frac{br}{a-1} \sum_i \alpha_i^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2 + br\sigma_{\alpha}^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2 + \frac{br}{a-1} \sum_i \alpha_i^2$
B	b - 1	$\sigma_e^2 + \frac{ar}{b-1} \sum_j \beta_j^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2 + ar\sigma_{\beta}^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2$
AB	(a-1)(b-1)	$\sigma_e^2 + \frac{r}{(a-1)(b-1)} \sum_i \sum_j \alpha\beta_{ij}^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2$
Error	ab(r-1)	σ_e^2	σ_e^2	σ_e^2

Fixed Model $Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk} \sim N(\mu + \alpha_i + \beta_j + \alpha\beta_{ij}, \sigma_e^2)$
 $i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r$
 $\varepsilon_{ijk} \sim N(0, \sigma_e^2) \quad \sum_i \alpha_i = \sum_j \beta_j = \sum_i \alpha\beta_{ij} = \sum_j \alpha\beta_{ij} = 0$

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Source	DF	MS	EMS Fixed
A Factor	a-1	MSA	$\sigma_e^2 + \frac{br}{a-1} \sum_i \alpha_i^2$
B Factor	b-1	MSB	$\sigma_e^2 + \frac{ar}{b-1} \sum_j \beta_j^2$
Interaction	(a-1)(b-1)	MSAB	$\sigma_e^2 + \frac{r}{(a-1)(b-1)} \sum_i \sum_j \alpha\beta_{ij}^2$
Error	ab(r-1)	MSE	σ_e^2

F test
 $H_0: \alpha_i = 0$ for all *i*; H_1 : not all $\alpha_i = 0$
 $F^* = MSA/MSE \sim F(a-1, ab(r-1))$ if H_0 is true
 H_0 : all $\beta_j = 0$; H_1 : not all $\beta_j = 0$
 $F^* = MSB/MSE \sim F(b-1, ab(r-1))$ if H_0 is true
 H_0 : all $\alpha\beta_{ij} = 0$; H_1 : not all $\alpha\beta_{ij} = 0$
 $F^* = MSAB/MSE \sim F((a-1)(b-1), ab(r-1))$ if H_0 is true

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Data Read;
Input Height \$ Width \$ Sales @@;
Datalines;
A1 B1 47 A1 B2 46
A1 B1 43 A1 B2 40
A2 B1 62 A2 B2 67
A2 B1 68 A2 B2 71
A3 B1 41 A3 B2 42
A3 B1 39 A3 B2 46

```

PROC GLM;
Class Height Width;
MODEL Sales = Height Width
Height*Width;
Means Height /LSD;
Means Height /Tukey;
Means Width /LSD;
Means Width /Tukey;
Means Height*Width /LSD;
Means Height*Width /Tukey;
RUN;
```

Dependent Variable: Sales

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	1580.00	316.00	30.58	0.0003
Height	2	1544.00	772.00	74.71	<.0001
Width	1	12.00	12.00	1.16	0.3226
Height*Width	2	24.00	12.00	1.16	0.3747
Error	6	62.00	10.33		

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t Grouping	Mean	N	Height
A	67.000	4	A2
B	44.000	4	A1
B	42.000	4	A3
Width			
A	52.000	6	B2
A	50.000	6	B1

Height	Width	N	Mean	Std Dev
A1	B1	2	45.0000000	2.82842712
A1	B2	2	43.0000000	4.24264069
A2	B1	2	65.0000000	4.24264069
A2	B2	2	69.0000000	2.82842712
A3	B1	2	44.0000000	2.82842712
A3	B2	2	44.0000000	2.82842712

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Case Study (2)

- ☐ Rice testing for variety and cultivation
- ☐ Factors:
 - Block (3 replications)
 - A: Varieties (A1 A2 A3)
 - B: Stands (108cm², 163cm², 218cm²)
 - 9 treatments (3 × 3)
- ☐ Experimental Unit: 27 plots (20m²) in 3 blocks
- ☐ Dependent Variable: yield (kg)

间距设置相同。

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处 理	区组 I	区组 II	区组 III	总和 T_{AB}
A_1B_1	8	8	8	24
A_1B_2	7	7	6	20
A_1B_3	6	5	6	17
A_2B_1	9	9	8	26
A_2B_2	7	9	6	22
A_2B_3	8	7	6	21
A_3B_1	7	7	6	20
A_3B_2	8	7	8	23
A_3B_3	10	9	9	28
总和 T	70	68	63	$T = 201$

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Data Rice;
Input Variety \$ Stand \$ Block Yield @@;
Datalines;

A1	B1	1	8	A1	B1	2	8	A1	B1	3	8
A1	B2	1	7	A1	B2	2	7	A1	B2	3	8
A1	B3	1	4	A1	B3	2	5	A1	B3	3	4
A2	B1	1	9	A2	B1	2	9	A2	B1	3	8
A2	B2	1	7	A2	B2	2	9	A2	B2	3	6
A2	B3	1	6	A2	B3	2	5	A2	B3	3	4
A3	B1	1	9	A3	B1	2	7	A3	B1	3	8
A3	B2	1	8	A3	B2	2	7	A3	B2	3	8
A3	B3	1	8	A3	B3	2	8	A3	B3	3	9

PROC GLM;
Class Variety Stand Block;
MODEL Yield = Variety Stand Variety*Stand
Block;
Means Variety /LSD;
Means Variety /Tukey;
Means Stand /LSD;
Means Stand /Tukey;
LSMeans Variety*Stand /STDERR PDIF;
RUN;

基于最小二乘的均值估计; STDERR:
打印出t检验($H_0: LSM=0$); PDIF:
打印出各平均数对比较后的统计显著程
度; TDIF: 打印出t值及显著程度

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Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	10	54.59259259	5.45925926	7.61	0.0002
Variety	2	9.85185185	4.92592593	6.86	0.0070
Stand	2	25.40740741	12.70370370	17.70	<.0001
Variety*Stand	4	18.81481481	4.70370370	6.55	0.0025
Block	2	0.51851852	0.25925926	0.36	0.7023
Error	16	11.48148148	0.71759259		

ANOVA Without Block Effects

Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	8	54.07407407	6.75925926	10.14	<.0001
Variety	2	9.85185185	4.92592593	7.39	0.0045
Stand	2	25.40740741	12.70370370	19.06	<.0001
Variety*Stand	4	18.81481481	4.70370370	7.06	0.0013
Error	18	12.00000000	0.66666667		

我查了数据

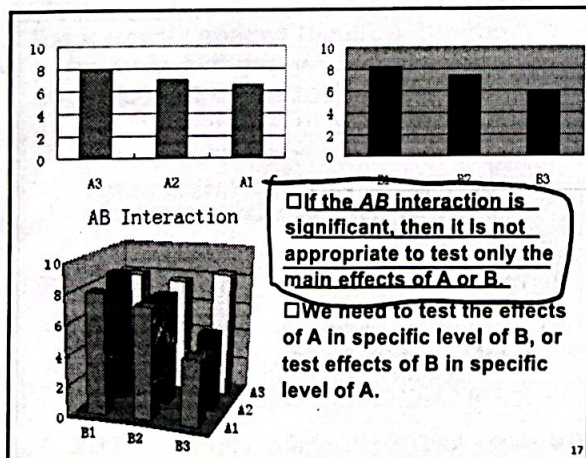
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Tukey Grouping Mean	N	Variety
A 8.0000	9	A3
B 7.0000	9	A2
B 6.5556	9	A1

Variety	Stand	N	Mean	Std Dev
A1	B1	3	8.00000000	0.00000000
A1	B2	3	7.33333333	0.57735027
A1	B3	3	4.33333333	0.57735027
A2	B1	3	8.66666667	0.57735027
A2	B2	3	7.33333333	1.52752523
A2	B3	3	5.00000000	1.00000000
A3	B1	3	8.00000000	1.00000000
A3	B2	3	7.66666667	0.57735027
A3	B3	3	8.33333333	0.57735027

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AB交互显著, 不能对A、B主效应下推断。



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1	A1	B1	8.00000000	1	A1	B1	8.00000000
2	A1	B2	7.33333333	4	A2	B1	8.66666667
3	A1	B3	4.33333333	7	A3	B1	8.00000000
4	A2	B1	8.66666667	2	A1	B2	7.33333333
5	A2	B2	7.33333333	5	A2	B2	7.33333333
6	A2	B3	5.00000000	8	A3	B2	7.66666667
7	A3	B1	8.00000000	3	A1	B3	4.33333333
8	A3	B2	7.66666667	6	A2	B3	5.00000000
9	A3	B3	8.33333333	9	A3	B3	8.33333333

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Least Squares Means for effect Variety*Stand
Pr > |t| for H0: LSMean(i) = LSMean(j)

ij	1	2	3	4	5	6	7	8	9
1		0.3495	<.0001	0.3495	0.3495	0.0005	1.0000	0.6364	0.6364
2	0.3495		0.0005	0.8718	1.0000	0.0039	0.3495	0.6364	0.1675
3	<.0001	0.0005		<.0001	0.0005	0.3495	<.0001	0.0002	<.0001
4	0.3495	0.0718	<.0001		0.0718	<.0001	0.3495	0.1675	0.6364
5	0.3495	1.0000	0.0005	0.8718		0.0039	0.3495	0.6364	0.1675
6	0.0005	0.0039	0.3495	<.0001	0.0039		0.0005	0.0014	0.0002
7	1.0000	0.3495	<.0001	0.3495	0.3495	0.0005		0.6364	0.6364
8	0.6364	0.6364	0.0002	0.1675	0.6364	0.0014	0.6364		0.3495
9	0.6364	0.1675	<.0001	0.6364	0.1675	0.0002	0.6364	0.3495	

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1, 2, 3 ; 4, 5, 6 ; 7, 8, 9

1, 4, 7 ; 2, 5, 8 ; 3, 6, 9

7, 8, 9 之间都不显著。

Analysis of Quantitative Factor

Factor:

- A is qualitative:
variety (A1 A2 A3)
- B is quantitative:
density (108cm², 163cm², 218cm²)

SAS command:

Class Variety Stand Block;
MODEL Yield = Variety|Stand Block;
Contrast 'Linear' Stand -1 0 1;
Contrast 'Quadratic' Stand 1 -2 1;

种植密度对产量的影响

是线性影响还是非线性影响。

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Data Rice;
Input Variety \$ Stand \$ Block Yield @@;
Datalines;

A1 B1 1 8 A1 B1 2 8 A1 B1 3 8
A1 B2 1 7 A1 B2 2 7 A1 B2 3 8
A1 B3 1 4 A1 B3 2 5 A1 B3 3 4
A2 B1 1 9 A2 B1 2 9 A2 B1 3 8
A2 B2 1 7 A2 B2 2 9 A2 B2 3 6
A2 B3 1 6 A2 B3 2 5 A2 B3 3 4
A3 B1 1 9 A3 B1 2 7 A3 B1 3 8
A3 B2 1 8 A3 B2 2 7 A3 B2 3 8
A3 B3 1 8 A3 B3 2 8 A3 B3 3 9

PROC GLM;
Class Variety Stand Block;
MODEL Yield = Variety|Stand Block;
Contrast 'Linear' Stand -1 0 1;
Contrast 'Quadratic' Stand 1 -2 1;
Means Variety /LSD;
RUN;

处理效应检验

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Source	DF	Sum Squares	Mean Square	F Value	Pr>F
Model	10	54.59259259	5.45925926	7.61	0.0002
Variety	2	9.85185185	4.92592593	6.86	0.0070
Stand	2	25.40740741	12.70370370	17.70	<.0001
Linear	1	24.50000000	24.50000000	34.14	<.0001
Quadratic	1	0.90740741	0.90740741	1.26	0.2774
Var*Sta	4	18.81481481	4.70370370	6.55	0.0025
Block	2	0.51851852	0.25925926	0.36	0.7023
Error	16	11.48148148	0.71759259		

The coefficients of linear effects and Quadratic effects is orthogonal, so the sum of squares of Stand is decomposed into two parts, one is due to the Linear effect and the other due to the Quadratic effect

密度对产量呈线性关系

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Contrast Test (df=1)	c ₁	c ₂	c ₃	Sum
Linear Contrast C1	-1	0	+1	0
Quadratic Contrast C2	+1	-2	+1	0
Orthogonal	-1	0	+1	0

Linear Contrast:

$$\tau_3 - \tau_1 = -1\tau_1 + 0\tau_2 + 1\tau_3$$

Quadratic Contrast:

$$(\tau_3 - \tau_2) - (\tau_2 - \tau_1) = 1\tau_1 - 2\tau_2 + 1\tau_3$$

Orthogonal:

$$(-1)(1) + (0)(-2) + (1)(1) = 0$$

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方差分析:

变异(平方和)分解, 剖析影响依变量变异的原因
比如, 上例的分析表明, 种植密度显著影响产量

问题:

实验只选择了3个种植密度, 其它种植密度如何影响产量? 即, 在其它种植密度情况下如何预测产量情况

回归分析: $E(y) = Xb$

可分析自变量对依变量的作用, 还可得到一回归模型, 对任意给定的一自变量可用此回归模型预测依变量
回归模型中一个自变量对应自由度为1

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$Y_{ijk} = \mu + \text{品种}_i + \text{密度}_j + \text{品种} \times \text{密度}_{ij} + \text{区组}_k + \varepsilon_{ijk}$

Linear effect: $\beta_1 X_{j(1)}$
Quadratic effect: $\beta_2 X_{j(2)}$

注：将数量型因子密度效应正交分解为1次型效应、2次型效应

$Y_{ijk} = \mu + \text{品种}_i + \beta_1 X_{j(1)} + \beta_2 X_{j(2)} + \text{品种} \times \text{密度}_{ij} + \text{区组}_k + \varepsilon_{ijk}$

注：互作效应也需做相应分解

密度3个水平	Level-1 108cm ² c_1	Level-2 163cm ² c_2	Level-3 218cm ² c_3	$\sqrt{\sum_{i=1}^3 c_i^2}$
Linear Contrast	-1	0	1	2
Quadratic Contrast	1	-2	1	6
$X_{j(1)}$	-1/√2	0	1/√2	1
$X_{j(2)}$	1/√6	-2/√6	1/√6	1

注：每一密度水平，两个系数代入模型

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$Y_{ijk} = \mu + \text{品种}_i + \beta_1 X_{j(1)} + \beta_2 X_{j(2)} + \text{品种} \times \text{密度}_{ij} + \text{区组}_k + \varepsilon_{ijk}$

密度3个水平	Level-1 108cm ² c_1	Level-2 163cm ² c_2	Level-3 218cm ² c_3	$c = \sqrt{\sum_{i=1}^3 c_i^2}$
Linear Contrast	-1	0	1	2
Quadratic Contrast	1	-2	1	6
$X_{j(1)}$	-1/√2	0	1/√2	1
$X_{j(2)}$	1/√6	-2/√6	1/√6	1

利用实验资料，可以估算上述模型的各项参数，如 β_1 、 β_2 ，从而获得一个回归模型，不妨记为：
 $\hat{Y} = \dots + \beta_1 X_1 + \beta_2 X_2 + \dots$

for any other plant density, such as 188, we can first calculate independent variable X_1 and X_2 , then predict $\hat{Y}$ using this regression equation

$X_1 = \frac{X_1(X - m)}{\sqrt{C(1)}} = \frac{1 \times (188 - 163)}{\sqrt{2}} = \frac{25}{\sqrt{2}}$

注： m 为上面3个水平的中值， Δ 为间距
(55 = 163 - 108 = 218 - 163)
 X_2 也有相应计算公式

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☆

3个水平 → 2个自由度 → 一次型/二次型效应

Predicting Y by the Xs

$Y_y = \beta_0 + [\beta_1 X_{1(1)} + \beta_2 X_{1(2)}] + \varepsilon_y$

where $X_i = c_i / \sqrt{\sum c_i^2}$ Effect of one level of the stand

Contrast Test (df = 1)	X_1	X_2	X_3	$\sqrt{\sum c_i^2}$
Linear	-1/√2	0	1/√2	2
Contrast C1	-1/√2	0	1/√2	2
Quadratic	1/√6	-2/√6	1/√6	6
Contrast C2	1/√6	-2/√6	1/√6	6

一次效应
二次效应

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Constructing Linear & Quadratic Xs

$X_i = c_i / \sqrt{C} = c_i / \sqrt{\sum c_i^2}$

Contrast Test (df = 1)	X_1	X_2	X_3
Linear Contrast XL	-1/√2	0	1/√2
Quadratic Contrast XQ	1/√6	-2/√6	1/√6

	Level-1	Level-2	Level-3
Contrast Test (df = 1)	X_1	X_2	X_3
Linear Contrast XL	-0.707107	0.000000	0.707107
Quadratic Contrast XQ	0.408248	-0.816497	0.408248

Above coefficients can be specified by the coefficient table of orthogonal decomposition for different number of levels

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for k evenly spaced levels, orthogonal polynomials of degree 1~4 can be conducted by:

$X_{(1)} = c_{(1)} / \sqrt{C_{(1)}} = \lambda_1 \left(\frac{x - m}{\Delta} \right) / \sqrt{C_{(1)}}$

$X_{(2)} = c_{(2)} / \sqrt{C_{(2)}} = \lambda_2 \left[\left(\frac{x - m}{\Delta} \right)^2 - \frac{k^2 - 1}{12} \right] / \sqrt{C_{(2)}}$

$X_{(3)} = c_{(3)} / \sqrt{C_{(3)}} = \lambda_3 \left[\left(\frac{x - m}{\Delta} \right)^3 - \frac{(x - m)(3k^2 - 7)}{20} \right] / \sqrt{C_{(3)}}$

$X_{(4)} = c_{(4)} / \sqrt{C_{(4)}} = \lambda_4 \left[\left(\frac{x - m}{\Delta} \right)^4 - \frac{(x - m)^2(3k^2 - 13)}{14} + \frac{3(k^2 - 1)(k^2 - 9)}{560} \right] / \sqrt{C_{(4)}}$

注：水平设置是等间距的，上面的 λ, C 可查相关正交分解系数表

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Level	$k=3$			$k=4$			$k=5$				
	P_1	P_2	P_3	P_1	P_2	P_3	P_4	P_1	P_2	P_3	P_4
1	-1	1	-3	1	-1	-2	2	-1	3	-1	1
2	0	-2	-1	-1	3	-1	-1	2	-4	2	-4
3	1	1	1	-1	-3	0	-2	0	6	0	6
4			3	1	1	1	-1	-2	-4	-2	-4
5						2	2	1	1	1	1
C	2	6	20	4	20	10	14	10	70	2	22
λ	1	3	2	1	$\frac{20}{3}$	1	$\frac{14}{3}$	1	$\frac{70}{3}$	1	$\frac{22}{3}$

Linear Quadratic Cubic quartic

Level	$k=6$					$k=7$					
	P_1	P_2	P_3	P_4	P_5	P_1	P_2	P_3	P_4	P_5	P_6
1	-5	5	-5	1	-1	-3	5	-1	3	-1	1
2	-3	-1	7	-3	5	-2	0	1	-7	4	-6
3	-1	-4	4	2	-10	-1	-3	1	1	-3	15
4	1	-4	-4	2	10	0	-4	0	6	0	-20
5	3	-1	-7	-3	-5	1	-3	-1	1	5	15
6	5	5	5	1	1	2	0	-1	-7	-4	-6
7						3	5	1	3	1	1
C	70	84	180	28	252	28	84	6	154	84	924
λ	2	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{15}$	$\frac{1}{6}$	1	1	$\frac{1}{2}$	$\frac{1}{15}$	$\frac{1}{35}$	$\frac{1}{105}$

k = number of levels, λ = constant, P_i = contrast vector for i th degree orthogonal polynomial
 C = squared length for P_i

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for k evenly spaced levels, orthogonal polynomials of degree 1~4 can be conducted by:

$$X_{(1)} = c_{(1)} / \sqrt{C_{(1)}} = \lambda_1 \left(\frac{x-m}{\Delta} \right) / \sqrt{C_{(1)}}$$

$$X_{(2)} = c_{(2)} / \sqrt{C_{(2)}} = \lambda_2 \left[\left(\frac{x-m}{\Delta} \right)^2 - \frac{k^2-1}{12} \right] / \sqrt{C_{(2)}}$$

$$X_{(3)} = c_{(3)} / \sqrt{C_{(3)}} = \lambda_3 \left[\left(\frac{x-m}{\Delta} \right)^3 - \frac{(k^2-7)}{20} \left(\frac{x-m}{\Delta} \right) \right] / \sqrt{C_{(3)}}$$

$$X_{(4)} = c_{(4)} / \sqrt{C_{(4)}} = \lambda_4 \left[\left(\frac{x-m}{\Delta} \right)^4 - \frac{(k^2-13)}{14} \left(\frac{x-m}{\Delta} \right)^2 + \frac{3(k^2-9)}{560} \right] / \sqrt{C_{(4)}}$$

Level	P_1	P_2	P_3	P_4
1	-1	1	-1	1
2	0	-2	1	-3
3	1	1	-1	1
4	2	6	3	1

$m = \frac{(108 + 163 + 218)}{3} = 163$
 $\Delta = 163 - 108 = 218 - 163 = 55$
 Linear, $\lambda = 1, C = 2$
 B1: 108cm²
 $X_{(1)} = 1 \times \frac{108-163}{\sqrt{2}} = -1/\sqrt{2}$

B2: 163cm²
 $X_{(1)} = 1 \times \frac{163-163}{\sqrt{2}} = 0$

B1: 218cm²
 $X_{(1)} = 1 \times \frac{218-163}{\sqrt{2}} = 1/\sqrt{2}$

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for k evenly spaced levels, orthogonal polynomials of degree 1~4 can be conducted by:

$$X_{(1)} = c_{(1)} / \sqrt{C_{(1)}} = \lambda_1 \left(\frac{x-m}{\Delta} \right) / \sqrt{C_{(1)}}$$

$$X_{(2)} = c_{(2)} / \sqrt{C_{(2)}} = \lambda_2 \left[\left(\frac{x-m}{\Delta} \right)^2 - \frac{k^2-1}{12} \right] / \sqrt{C_{(2)}}$$

$$X_{(3)} = c_{(3)} / \sqrt{C_{(3)}} = \lambda_3 \left[\left(\frac{x-m}{\Delta} \right)^3 - \frac{(k^2-7)}{20} \left(\frac{x-m}{\Delta} \right) \right] / \sqrt{C_{(3)}}$$

$$X_{(4)} = c_{(4)} / \sqrt{C_{(4)}} = \lambda_4 \left[\left(\frac{x-m}{\Delta} \right)^4 - \frac{(k^2-13)}{14} \left(\frac{x-m}{\Delta} \right)^2 + \frac{3(k^2-9)}{560} \right] / \sqrt{C_{(4)}}$$

Level	P_1	P_2	P_3	P_4
1	-1	1	-1	1
2	0	-2	1	-3
3	1	1	-1	1
4	2	6	3	1

Quadratic, $\lambda = 3, C = 6$
 B1: 108cm²
 $X_{(2)} = 3 \times \left[\left(\frac{108-163}{55} \right)^2 - \frac{(9-1)}{12} \right] / \sqrt{6} = 1/\sqrt{6}$

B2: 163cm²
 $X_{(2)} = 3 \times \left[\left(\frac{163-163}{55} \right)^2 - \frac{(9-1)}{12} \right] / \sqrt{6} = -2/\sqrt{6}$

B1: 218cm²
 $X_{(2)} = 3 \times \left[\left(\frac{218-163}{55} \right)^2 - \frac{(9-1)}{12} \right] / \sqrt{6} = 1/\sqrt{6}$

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Contrast Test (df=1)	c_1	c_2	c_3	c_4	Sum
Linear Contrast C1	-3	-1	+1	+3	0
Quadratic Contrast C2	+1	-1	-1	+1	0
Cubic Contrast C3	-1	+3	-3	+1	0

Linear Contrast:
 $-3\tau_1 - 1\tau_2 + 1\tau_3 + 3\tau_4$

Quadratic Contrast:
 $1\tau_1 - 1\tau_2 - 1\tau_3 + 1\tau_4$

Cubic Contrast:
 $-1\tau_1 + 3\tau_2 - 3\tau_3 + 1\tau_4$

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Predicting Y by the Xs

$$Y_{ij} = \beta_0 + \beta_1 X_{i(1)} + \beta_2 X_{i(2)} + \beta_3 X_{i(3)} + \epsilon_{ij}$$

where $X_i = c_i / \sqrt{\sum c_i^2}$

Contrast Test (df=1)	X_1	X_2	X_3	X_4
Linear Contrast C1	$-3/\sqrt{20}$	$-1/\sqrt{20}$	$1/\sqrt{20}$	$3/\sqrt{20}$
Quadratic Contrast C2	$1/\sqrt{4}$	$-1/\sqrt{4}$	$-1/\sqrt{4}$	$1/\sqrt{4}$
Cubic Contrast C3	$-1/\sqrt{20}$	$3/\sqrt{20}$	$-3/\sqrt{20}$	$1/\sqrt{20}$

Contrast Test (df=1)	X_1	X_2	X_3	X_4
Linear Contrast C1	0.6708	-0.2236	0.2236	0.6708
Quadratic Contrast C2	0.5000	-0.5000	-0.5000	0.5000
Cubic Contrast C3	-0.2236	0.6708	-0.6708	0.2236

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For case (3), factor has four levels:
 6, 8, 10, and 12 training hours

$m = 9, \Delta = 2, \lambda_1 = 2, \lambda_2 = 1, \lambda_3 = 10/3,$
 $C_1 = 20, C_2 = 4, C_3 = 20$

For x hours,

$$X_{(1)} = c_{(1)} / \sqrt{C_{(1)}} = \lambda_1 \left(\frac{x-m}{\Delta} \right) / \sqrt{C_{(1)}}$$

$$X_{(2)} = c_{(2)} / \sqrt{C_{(2)}} = \lambda_2 \left[\left(\frac{x-m}{\Delta} \right)^2 - \frac{k^2-1}{12} \right] / \sqrt{C_{(2)}}$$

$$X_{(3)} = c_{(3)} / \sqrt{C_{(3)}} = \lambda_3 \left[\left(\frac{x-m}{\Delta} \right)^3 - \frac{(k^2-7)}{20} \left(\frac{x-m}{\Delta} \right) \right] / \sqrt{C_{(3)}}$$

Level	P_1	P_2	P_3
1	-3	1	-1
2	-1	-1	3
3	1	-1	-3
4	3	1	1

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Data Rice;
 Input Variety \$ Stand \$ Block Yield @@;
 Datalines;

A1	B1	1	8	A1	B1	2	8	A1	B1	3	8
A1	B2	1	7	A1	B2	2	7	A1	B2	3	8
A1	B3	1	4	A1	B3	2	5	A1	B3	3	4
A2	B1	1	9	A2	B1	2	9	A2	B1	3	8
A2	B2	1	7	A2	B2	2	9	A2	B2	3	6
A2	B3	1	6	A2	B3	2	5	A2	B3	3	4
A3	B1	1	9	A3	B1	2	7	A3	B1	3	8
A3	B2	1	8	A3	B2	2	7	A3	B2	3	8
A3	B3	1	8	A3	B3	2	8	A3	B3	3	9

B1 : replaced with 0.70107
 B2 : replaced with 0.00000
 B3 : replaced with 0.70107

Coefficients for linear effect

0.408248
 -0.816497
 0.408248

Coefficients for quadratic effect

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XL XQ

```

Data Rice;
Input Variety $ Block XL XQ Yield @@;
Datalines;
A1 1 -0.707107 0.408248 8 A1 2 -0.707107 0.408248 8 A1 3 -0.707107 0.408248 8
A1 1 0.000000 -0.816497 7 A1 2 0.000000 -0.816497 7 A1 3 0.000000 -0.816497 8
A1 1 -0.707107 0.408248 4 A1 2 0.707107 0.408248 5 A1 3 0.707107 0.408248 4
A2 1 -0.707107 0.408248 9 A2 2 -0.707107 0.408248 9 A2 3 -0.707107 0.408248 8
A2 1 0.000000 -0.816497 7 A2 2 0.000000 -0.816497 9 A2 3 0.000000 -0.816497 8
A2 1 0.707107 0.408248 6 A2 2 0.707107 0.408248 5 A2 3 0.707107 0.408248 4
A3 1 -0.707107 0.408248 9 A3 2 -0.707107 0.408248 7 A3 3 -0.707107 0.408248 8
A3 1 0.000000 -0.816497 8 A3 2 0.000000 -0.816497 7 A3 3 0.000000 -0.816497 8
A3 1 0.707107 0.408248 8 A3 2 0.707107 0.408248 8 A3 3 0.707107 0.408248 9
;
PROC GLM;
Class Variety Block;
MODEL Yield = Variety XL XQ
        Variety*XL Variety*XQ Block;
Means Variety /LSD;
RUN;

```

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Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	10	54.59259259	5.45925926	7.61	0.0002
Variety	2	9.85185185	4.92592593	6.86	0.0070
XL	1	24.50000000	24.50000000	34.14	<.0001
XQ	1	0.90740741	0.90740741	1.26	0.2774
XL*Var	2	16.00000000	8.00000000	11.15	0.0009
XQ*Var	2	2.81481481	1.40740741	1.96	0.1731
Block	2	0.51851852	0.25925926	0.36	0.7023
Error	16	11.48148148	0.71759259		

Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	10	54.59259259	5.45925926	7.61	0.0002
Variety	2	9.85185185	4.92592593	6.86	0.0070
Stand	2	25.40740741	12.70370370	17.70	<.0001
Linear	1	24.50000000	24.50000000	34.14	<.0001
Quadratic	1	0.90740741	0.90740741	1.26	0.2774
Var*Sta	4	18.81481481	4.70370370	6.55	0.0025
Block	2	0.51851852	0.25925926	0.36	0.7023
Error	16	11.48148148	0.71759259		

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Regression approach to two-way factorial designs

$$Y = b_0 + b_1X_1 + b_2X_2 + \dots + b_pX_p + \varepsilon$$

Effect coding

□ Determine the variables for A factor

Contrast (df = 2)	A ₁	A ₂	A ₃	A ₃	Dummy coding
X1	1	0	-1	0	两种编码方式
X2	0	1	-1	0	

1 if observation from level 1 for factor A
 $X_1 = -1$ if observation from level 3 for factor A
 0 otherwise

1 if observation from level 2 for factor A
 $X_2 = -1$ if observation from level 3 for factor A
 0 otherwise

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□ Determine the variables for quantitative B factor

Contrast (df = 2)	B ₁	B ₂	B ₃	$\sqrt{\sum c_i^2}$
X3(Linear)	$-1/\sqrt{2}$	0	$1/\sqrt{2}$	2
X4(Quadratic)	$1/\sqrt{6}$	$-2/\sqrt{6}$	$1/\sqrt{6}$	6

□ Determine the variables for AB interaction factor

□ $X_1 \times X_3, X_2 \times X_3$ for A × Linear

□ $X_1 \times X_4, X_2 \times X_4$ for A × Quadratic

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X1	X2	X3	X4	Y	X1	X2	X3	X4	Y	X1	X2	X3	X4	Y
1	0	-0.707107	0.408248	8	1	0	-0.707107	0.408248	8	1	0	-0.707107	0.408248	8
1	0	0.000000	-0.816497	7	1	0	0.000000	-0.816497	7	1	0	0.000000	-0.816497	8
1	0	0.707107	0.408248	4	1	0	0.707107	0.408248	5	1	0	0.707107	0.408248	4
0	1	-0.707107	0.408248	9	0	1	-0.707107	0.408248	9	0	1	-0.707107	0.408248	8
0	1	0.000000	-0.816497	7	0	1	0.000000	-0.816497	9	0	1	0.000000	-0.816497	6
0	1	0.707107	0.408248	6	0	1	0.707107	0.408248	5	0	1	0.707107	0.408248	4
-1	-1	-0.707107	0.408248	9	-1	-1	-0.707107	0.408248	7	-1	-1	-0.707107	0.408248	8
-1	-1	0.000000	-0.816497	8	-1	-1	0.000000	-0.816497	7	-1	-1	0.000000	-0.816497	8
-1	-1	0.707107	0.408248	8	-1	-1	0.707107	0.408248	8	-1	-1	0.707107	0.408248	9

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Codes for variety

```

Data Rice;
Input X1 X2 X3 X4 Block Yield @@;
A1L = X1 * X3;
A2L = X2 * X3;
Datalines;
1 0 -0.707107 0.408248 1 8 1 0 -0.707107 0.408248 2 8 1 0 -0.707107 0.408248 3 8
1 0 0.000000 -0.816497 1 7 1 0 0.000000 -0.816497 2 7 1 0 0.000000 -0.816497 3 8
1 0 0.707107 0.408248 1 4 1 0 0.707107 0.408248 2 5 1 0 0.707107 0.408248 3 4
0 1 -0.707107 0.408248 1 9 0 1 -0.707107 0.408248 2 9 0 1 -0.707107 0.408248 3 8
0 1 0.000000 -0.816497 1 7 0 1 0.000000 -0.816497 2 9 0 1 0.000000 -0.816497 3 6
0 1 0.707107 0.408248 1 6 0 1 0.707107 0.408248 2 5 0 1 0.707107 0.408248 3 4
-1 -1 -0.707107 0.408248 1 9 -1 -1 -0.707107 0.408248 2 7 -1 -1 -0.707107 0.408248 3 8
-1 -1 0.000000 -0.816497 1 8 -1 -1 0.000000 -0.816497 2 7 -1 -1 0.000000 -0.816497 3 8
-1 -1 0.707107 0.408248 1 8 -1 -1 0.707107 0.408248 2 8 -1 -1 0.707107 0.408248 3 9
;
PROC REG;
MODEL Yield = X1 X2 X3 X4 A1L A2L;
RUN;

```

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品种 (A2显著)

线性关系

2个显著

Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	6	51.25925926	8.54320988	11.53	<.0001
X1	1	9.38888889	9.38888889	12.68	0.0020
X2	1	0.46296296	0.46296296	0.62	0.4385
X3	1	24.50000000	24.50000000	33.08	<.0001
X4	1	0.90740741	0.90740741	1.23	0.2815
A1L	1	12.00000000	12.00000000	16.20	0.0007
A2L	1	4.00000000	4.00000000	5.40	0.0308
Error	20	14.81481481	0.74074074		

Parameter	Estimate	SE	t Value	Pr > t
Intercept	7.185185079	0.16563467	43.38	<.0001
X1	-0.629629630	0.23424279	-2.69	0.0141
X2	-0.185185185	0.23424279	-0.79	0.4385
X3	-1.649915312	0.28688757	-5.75	<.0001
X4	-0.317526415	0.28688763	-1.11	0.2815
A1L	-0.942808750	0.40572029	-2.32	0.0308
A2L	-0.942808750	0.40572029	-2.32	0.0308

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Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	4	49.88888889	12.47222222	16.95	<.0001
X1	1	9.38888889	9.38888889	12.76	0.0017
X3	1	24.50000000	24.50000000	33.30	<.0001
A1L	1	4.00000000	4.00000000	5.44	0.0293
A2L	1	4.00000000	4.00000000	5.44	0.0293
Error	22	16.18518519	0.73569024		

Parameter	Estimate	SE	t Value	Pr > t
Intercept	7.185185185	0.16506904	43.53	<.0001
X1	-0.722222222	0.20216745	-3.57	0.0017
X3	-1.649915312	0.28590787	-5.77	<.0001
A1L	-0.942808750	0.40433478	-2.33	0.0293
A2L	-0.942808750	0.40433478	-2.33	0.0293

$\hat{Y} = 7.185 - 0.722X_1 - 1.650X_3 - 0.943X_1X_3 - 0.943X_2X_3$
 ? What is the X_3

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□ Three levels of factor B: 108cm², 163cm², 218cm².

$m = 163, \Delta = 55, \lambda_1 = 1, C_1 = 2$

□ At density of seeding of 200cm², Real value of one stand

$X_3 = \frac{(200 - 163)}{55} / \sqrt{2};$

□ Predicting yield Coefficient for the prediction model

$\hat{Y} = 7.185 - 0.722X_1 - 1.650X_3 - 0.943X_1X_3 - 0.943X_2X_3$

How to obtain the prediction model for A1, A2, A3, respectively

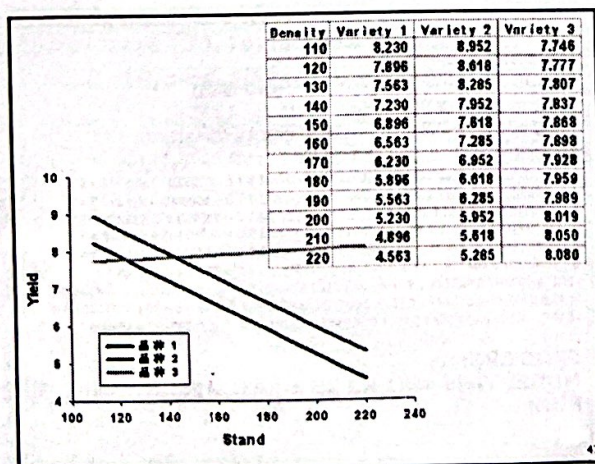
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$X_3 = \frac{(B - 163)}{55} / \sqrt{2}$

A	B	X1	X2	X3	X1X3	X2X3	Y
A1	110	1	0	-0.681	-0.681	0.000	8.230
A1	120	1	0	-0.553	-0.553	0.000	7.896
A1	130	1	0	-0.424	-0.424	0.000	7.563
A1	140	1	0	-0.296	-0.296	0.000	7.230
A1	150	1	0	-0.167	-0.167	0.000	6.896
A1	160	1	0	-0.039	-0.039	0.000	6.563
A1	170	1	0	0.090	0.090	0.000	6.230
A1	180	1	0	0.219	0.219	0.000	5.896
A1	190	1	0	0.347	0.347	0.000	5.563
A1	200	1	0	0.476	0.476	0.000	5.230
A1	210	1	0	0.604	0.604	0.000	4.896
A1	220	1	0	0.733	0.733	0.000	4.563

$\hat{Y} = 7.185 - 0.722X_1 - 1.650X_3 - 0.943X_1X_3 - 0.943X_2X_3$

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Random Model

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk}$$

$$\sim N(\mu, \sigma_\varepsilon^2 + \sigma_{\alpha\beta}^2 + \sigma_\beta^2 + \sigma_\alpha^2)$$

$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r;$

$$\varepsilon_{ijk} \sim N(0, \sigma_\varepsilon^2), \quad \alpha_i \sim N(0, \sigma_\alpha^2)$$

$$\beta_j \sim N(0, \sigma_\beta^2), \quad \alpha\beta_{ij} \sim N(0, \sigma_{\alpha\beta}^2)$$

→ EMS(A), EMS(B), EMS(AB), EMS(Error)

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Source	df	MS	E(MS) Random
Factor A	$a-1$	MSA	$\sigma_e^2 + r\sigma_{\alpha}^2 + br\sigma_{\alpha}^2$
Factor B	$b-1$	MSB	$\sigma_e^2 + r\sigma_{\beta}^2 + ar\sigma_{\beta}^2$
Interaction	$(a-1)(b-1)$	$MSAB$	$\sigma_e^2 + r\sigma_{\alpha\beta}^2$
Error	$ab(r-1)$	MSE	σ_e^2

F test
 $H_0: \sigma_{\alpha}^2 = 0; H_1: \sigma_{\alpha}^2 > 0$
 $F = MSA/MSAB \sim F(a-1, (a-1)(b-1))$ if H_0 is true
 $H_0: \sigma_{\beta}^2 = 0; H_1: \sigma_{\beta}^2 > 0$
 $F = MSB/MSAB \sim F(b-1, (a-1)(b-1))$ if H_0 is true
 $H_0: \sigma_{\alpha\beta}^2 = 0; H_1: \sigma_{\alpha\beta}^2 > 0$
 $F = MSAB/MSE \sim F((a-1)(b-1), ab(r-1))$ if H_0 is true

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Case Study (2)	
☐	Rice testing for variety and cultivation
☐	Factors (Random):
☐	Block (3 replications)
☐	A: Varieties (A1 A2 A3)
☐	B: Stands (108cm ² , 163cm ² , 218cm ²)
☐	9 treatments (3 × 3)
☐	Experimental Unit: 27 plots (20m ²) in 3 blocks
☐	Dependent Variable: yield (kg)

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Data Rice;	
Input Variety \$ Stand \$ Block Yield @@;	
Datalines;	
A1 B1 1 8 A1 B1 2 8 A1 B1 3 8	Random: declare random variables in
A1 B2 1 7 A1 B2 2 7 A1 B2 3 8	"model statement", calculate expected
A1 B3 1 4 A1 B3 2 5 A1 B3 3 4	values of mean squares (EMS), optionally,
A2 B1 1 9 A2 B1 2 9 A2 B1 3 8	to perform random effects analysis of
A2 B2 1 7 A2 B2 2 9 A2 B2 3 6	variance tests in F-statistic with
A2 B3 1 6 A2 B3 2 5 A2 B3 3 4	denominator determined by the EMS
A3 B1 1 9 A3 B1 2 7 A3 B1 3 8	Although an F value is computed for all sums of
A3 B2 1 8 A3 B2 2 7 A3 B2 3 8	squares in the analysis using the residual MS as
A3 B3 1 8 A3 B3 2 8 A3 B3 3 9	an error term, you may request additional F
	tests using other effects as error terms. You
	need a TEST statement when a nonstandard
	error structure (as in a split-plot design) exists
	specifies one, and only one, effect to
	use as the error (denominator) term
PROC GLM data=Rice;	
Class Variety Stand Block;	
MODEL Yield = Variety Stand Variety*Stand Block;	
Random Variety Stand Variety*Stand Block/test;	
Test H = Variety Stand E = Variety*Stand;	
RUN;	
PROC VARCOMP method=REML;	
Class Variety Stand Block;	
MODEL Yield = Variety Stand Variety*Stand Block;	
RUN;	

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Not Selecting for Error MS					
Source	DF	Type I SS	Mean Square	F Value	Pr > F
Variety	2	9.85185185	4.92592593	6.86	0.0070
Stand	2	25.40740741	12.70370370	17.70	<.0001
Variety*Stand	4	18.81481481	4.70370370	6.55	0.0025
Block	2	0.51851852	0.25925926	0.36	0.7023

Source DF Type III SS Mean Square F Value Pr > F					
Source	DF	Type I SS	Mean Square	F Value	Pr > F
Variety	2	9.851852	4.925926	1.05	0.4308
Stand	2	25.407407	12.703704	2.70	0.1810
Error(V×S)	4	18.814815	4.703704		

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REML Estimates		
Variance	Estimate	SE
Var(Variety)	0.02469	0.66041
Var(Stand)	0.88889	1.45910
Var(Variety*Stand)	1.34568	1.11115
Var(Block)	0	0.00000
Var(Error)	0.66667	0.22222

Asymptotic Covariance Matrix of Estimates					
	Variety	Stand	Var*Sta	Block	Error
Variety	0.43614	0.13657	-0.40972	0.00000	0.00000
Stand	0.13657	2.12898	-0.40972	0.00000	0.00000
Var*Sta	-0.40972	-0.40972	1.23465	0.00000	-0.01646
Block	0.00000	0.00000	0.00000	0.00000	0.00000
Error	0.00000	0.00000	-0.01646	0.00000	0.04938

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Mixed Model	
$Y_{ijk} = \mu + \alpha_i + \beta_j + \alpha\beta_{ij} + \varepsilon_{ijk}$	
$\sim N(\mu + \alpha_i, \sigma_e^2 + \sigma_{\alpha}^2 + \sigma_{\alpha\beta}^2 + \sigma_{\beta}^2)$	
$i = 1, 2, \dots, a; j = 1, 2, \dots, b; k = 1, 2, \dots, r;$	
$\sum_i \alpha_i = 0$	
$\beta_j \sim N(0, \sigma_{\beta}^2), \quad \alpha\beta_{ij} \sim N(0, \sigma_{\alpha\beta}^2)$	
$\varepsilon_{ijk} \sim N(0, \sigma_e^2)$	

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Source	df	MS	E(MS) Mixed
Factor A	a-1	MSA	$\frac{\sigma_e^2 + r\sigma_{\alpha\beta}^2}{a-1} + \frac{br}{a-1} \sum \alpha_i^2$
Factor B	b-1	MSB	$\frac{\sigma_e^2 + r\sigma_{\alpha\beta}^2}{b-1} + \frac{ar}{b-1} \sum \beta_j^2$
Interaction (a-1)(b-1)	MSAB	$\frac{\sigma_e^2 + r\sigma_{\alpha\beta}^2}{(a-1)(b-1)}$	
Error	ab(r-1)	MSE	σ_e^2

F test
 $H_0: \sum \alpha_i^2 = 0; H_1: \sum \alpha_i^2 > 0$
 $F^* = MSA/MSAB \sim F(a-1, (a-1)(b-1))$ if H_0 is true
 $H_0: \sigma_{\alpha\beta}^2 = 0; H_1: \sigma_{\alpha\beta}^2 > 0$
 $F^* = MSB/MSE \sim F(b-1, ab(r-1))$ if H_0 is true
 $H_0: \sigma_{\alpha\beta}^2 = 0; H_1: \sigma_{\alpha\beta}^2 > 0$
 $F^* = MSAB/MSE \sim F((a-1)(b-1), ab(r-1))$ if H_0 is true

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Case Study (2)
☐ Rice testing for variety and cultivation
☐ Factors (Mixed):
■ Block (3 replications)
■ A: Varieties (A1 A2 A3)
■ B: Stands (108cm ² , 163cm ² , 218cm ²)
■ 9 treatments (3 × 3)
☐ Experimental Unit: 27 plots (20m ²) in 3 blocks
☐ Dependent Variable: yield (kg)

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Data Rice;
Input Variety \$ Stand \$ Block Yield @@;
Datalines;
A1 B1 1 8 A1 B1 2 8 A1 B1 3 8
A1 B2 1 7 A1 B2 2 7 A1 B2 3 8
A1 B3 1 4 A1 B3 2 5 A1 B3 3 4
A2 B1 1 9 A2 B1 2 9 A2 B1 3 8
A2 B2 1 7 A2 B2 2 9 A2 B2 3 6
A2 B3 1 6 A2 B3 2 5 A2 B3 3 4
A3 B1 1 9 A3 B1 2 7 A3 B1 3 8
A3 B2 1 8 A3 B2 2 7 A3 B2 3 8
A3 B3 1 8 A3 B3 2 8 A3 B3 3 9
;

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PROC GLM data=Rice;
Class Variety Stand Block;
MODEL Yield = Variety Stand Block;
Random Stand Variety*Stand Block/test;
Test H = Variety E = Variety*Stand;
LSMeans Variety/ STDERR PDIF ADJUST=TUKEY E =
Variety*Stand;
RUN;

In contrast to the MEANS statement, the LSMEANS statement performs multiple comparisons on interactions as well as main effects. Adjust: request a multiple comparison adjustment for the p-value and confidence limits for the difference of LS-Mean. Adjust=... the option changes the results of TDIF and PDIF.

PROC MIXED data=Rice method=type3;
Class Variety Stand Block;
MODEL Yield = Variety/SOLUTION;
Random Stand Variety*Stand Block/SOLUTION;
LSMeans Variety/ ADJUST=TUKEY;
RUN;

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Results of PROC GLM
Not Selecting for Error MS
Source DF Type I SS Mean Square F Value Pr > F
Variety 2 9.85185185 4.92592593 6.86 0.0070
Stand 2 25.40740741 12.70370370 17.70 <.0001
Variety*Stand 4 18.81481481 4.70370370 6.55 0.0025
Block 2 0.51851852 0.25925926 0.36 0.7023

Source DF Type III SS Mean Square F Value Pr > F
Variety 2 9.851852 4.925926 1.05 0.4308
Stand 2 25.40740741 12.70370370 17.70 <.0001
Error(VXS) 4 18.814815 4.703704

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Source DF Type I SS Mean Square F Value Pr > F
Variety 2 9.85185185 4.92592593 6.86 0.0070
Stand 2 25.40740741 12.70370370 17.70 <.0001
Variety*Stand 4 18.81481481 4.70370370 6.55 0.0025
Block 2 0.51851852 0.25925926 0.36 0.7023
Variety 2 9.851852 4.925926 1.05 0.4308
Error 4 18.814815 4.703704

Source Type III Expected Mean Square
Variety Var(Error) + 3Var(Variety*Stand) + Q(Variety)
Stand Var(Error) + 3Var(Variety*Stand) + 9Var(Stand)
Variety*Stand Var(Error) + 3Var(Variety*Stand)
Block Var(Error) + 9Var(Block)

The expected value of mean squares of Stand given in SAS is wrong, the red item in the above table should not be included

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Standard LSMEAN				
Variety	LSMEAN	SE	Pr > t	Number
A1	6.55555556	0.72293412	0.0008	1
A2	7.00000000	0.72293412	0.0006	2
A3	8.00000000	0.72293412	0.0004	3

Least Squares Means for effect Variety
Pr > |t| for H0: LSMean(i)=LSMean(j)

i/j	1	2	3
1		0.9035	0.4173
2	0.9035		0.6268
3	0.4173	0.6268	

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Results of PROC MIXED				
Source	DF	SS	MS	Expected Mean Square
Variety	2	9.851852	4.925926	Var(Res)+3Var(Var*Sta)+0(Var(Variety))
Stand	2	25.407407	12.703704	Var(Res)+3Var(Var*Sta)+9Var(Stand)
Var*Sta	4	18.814815	4.703704	Var(Res)+3Var(Var*Sta)
Block	2	0.518519	0.259259	Var(Res)+9Var(Block)
Residual	16	11.481481	0.717593	Var(Res)

Source	Error Term	DF	F	Pr > F Value
Variety	MS(Var*Sta)	4	1.05	0.4308
Stand	MS(Var*Sta)	4	2.70	0.1310
Var*Sta	MS(Residual)	16	6.55	0.0025
Block	MS(Residual)	16	0.36	0.7023

Cov Parm	Estimate
Stand	0.8889
Var*Sta	1.3287
Block	-0.05093
Residual	0.7176

The incorrect denominator is used in calculating F statistic when performing significance testing for Stand effect

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Solution for Fixed Effects						
Effect	Variety	Estimate	SE	DF	t Value	Pr > t
Intercept		8.0000	0.8955	2	8.93	0.0123
Variety	A1	-1.4444	1.0224	4	-1.41	0.2306
Variety	A2	-1.0000	1.0224	4	-0.98	0.3834
Variety	A3	0				

Solution for Random Effects						
Effect	Variety	Stand	Block	DF	t Value	Pr > t
Stand		B1		16	0.91	0.3766
Stand		B2		16	0.23	0.8230
Stand		B3		16	-1.14	0.2724

Effect	Variety	Stand	Block	DF	t Value	Pr > t
Var*Sta	A1	B1		16	0.78	0.4457
Var*Sta	A1	B2		16	0.61	0.5523
Var*Sta	A1	B3		16	-1.39	0.1838
Var*Sta	A2	B1		16	1.00	0.3315
Var*Sta	A2	B2		16	0.17	0.8687
Var*Sta	A2	B3		16	-1.17	0.2593
Var*Sta	A3	B1		16	-0.65	0.5279
Var*Sta	A3	B2		16	-0.49	0.6383
Var*Sta	A3	B3		16	1.14	0.2727
Block			1	16	-Infy	<.0001
Block			2	16	-Infy	<.0001
Block			3	16	-Infy	<.0001

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7.2 Multi-Way Factorial Designs

□ A study of the effects of sex, body fat and smoking history on exercise tolerance in stress testing of persons 25 to 35 years old.

□ Factor:

- A: sex (male, female)
- B: Body Fat (low, high)
- C: Smoking history (light, heavy)
- 8 treatments (2 × 2 × 2)

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Smoking History		
	k = 1 Light	k = 2 Heavy
j = 1 Low fat:		
i = 1 Male	24 (Y ₁₁₁₁)	18 (Y ₁₁₂₁)
	29 (Y ₁₁₁₂)	19 (Y ₁₁₂₂)
	25 (Y ₁₁₁₃)	23 (Y ₁₁₂₃)
i = 2 Female	20 (Y ₂₁₁₁)	15 (Y ₂₁₂₁)
	22 (Y ₂₁₁₂)	10 (Y ₂₁₂₂)
	18 (Y ₂₁₁₃)	11 (Y ₂₁₂₃)
j = 2 High fat:		
i = 1 Male	15 (Y ₁₂₁₁)	15 (Y ₁₂₂₁)
	15 (Y ₁₂₁₂)	20 (Y ₁₂₂₂)
	12 (Y ₁₂₁₃)	13 (Y ₁₂₂₃)
i = 2 Female	16 (Y ₂₂₁₁)	10 (Y ₂₂₂₁)
	9 (Y ₂₂₁₂)	14 (Y ₂₂₂₂)
	11 (Y ₂₂₁₃)	6 (Y ₂₂₂₃)

$$Y_{ijkm} = \mu + \alpha_i + \beta_j + \gamma_k + \alpha\beta_{ij} + \alpha\gamma_{ik} + \beta\gamma_{jk} + \alpha\beta\gamma_{ijk} + \epsilon_{ijkm}$$

i = 1, 2, ..., a; j = 1, 2, ..., b; k = 1, 2, ..., c; m = 1, 2, ..., r

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二阶互作 + 三阶互作

Partition of of total sum of squares and degrees of freedom

$$\sum_i \sum_j \sum_k \sum_m (Y_{ijkm} - \bar{Y}_{....})^2 =$$

$$bcr \sum_i (\bar{Y}_{i...} - \bar{Y}_{....})^2 + acr \sum_j (\bar{Y}_{.j.} - \bar{Y}_{....})^2 + abr \sum_k (\bar{Y}_{...k} - \bar{Y}_{....})^2$$

$$+ cr \sum_i \sum_j (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j.} + \bar{Y}_{....})^2 + br \sum_i \sum_k (\bar{Y}_{i.k.} - \bar{Y}_{i...} - \bar{Y}_{...k} + \bar{Y}_{....})^2$$

$$+ ar \sum_j \sum_k (\bar{Y}_{.jk.} - \bar{Y}_{.j.} - \bar{Y}_{...k} + \bar{Y}_{....})^2$$

$$+ r \sum_i \sum_j \sum_k (\bar{Y}_{ijk.} - \bar{Y}_{ij..} - \bar{Y}_{i.k.} - \bar{Y}_{.jk.} + \bar{Y}_{i...} + \bar{Y}_{.j.} + \bar{Y}_{...k} - \bar{Y}_{....})^2 + SSE$$

$$SSTO = SSA + SSB + SSC + SSAB + SSAC + SSBC + SSABC + SSE$$

$$df_{TO} = df_A + df_B + df_C + df_{AB} + df_{AC} + df_{BC} + df_{ABC} + df_r$$

$$abr - 1 = (a-1) + (b-1) + (c-1) + (a-1)(b-1) + (a-1)(c-1) + (b-1)(c-1) + (a-1)(b-1)(c-1) + abc(r-1)$$

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ANOVA table (fixed model)			
Source	df	E(MS) Fixed	F*
A	a-1	$\sigma^2 + \frac{bcr}{a-1} \sum \alpha_i^2$	MSA/MSE
B	b-1	$\sigma^2 + \frac{acr}{b-1} \sum \beta_j^2$	MSB/MSE
C	c-1	$\sigma^2 + \frac{abr}{c-1} \sum \gamma_k^2$	MSC/MSE
AB	(a-1)(b-1)	$\sigma^2 + \frac{cr}{(a-1)(b-1)} \sum \alpha\beta_{ij}^2$	MSAB/MSE
AC	(a-1)(c-1)	$\sigma^2 + \frac{br}{(a-1)(c-1)} \sum \alpha\gamma_{ik}^2$	MSAC/MSE
BC	(b-1)(c-1)	$\sigma^2 + \frac{ar}{(b-1)(c-1)} \sum \beta\gamma_{jk}^2$	MSBC/MSE
ABC	(a-1)(b-1)(c-1)	$\sigma^2 + \frac{r}{(a-1)(b-1)(c-1)} \sum \alpha\beta\gamma_{ijk}^2$	
Error	abc(r-1)	σ^2	

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F test

H_0 : all $\alpha_i = 0$; H_1 : not all $\alpha_i = 0$

$F^* = MSA/MSE \sim F(a-1, abc(r-1))$ if H_0 is true

H_0 : all $\beta_j = 0$; H_1 : not all $\beta_j = 0$

$F^* = MSB/MSE \sim F(b-1, abc(r-1))$ if H_0 is true

H_0 : all $\gamma_k = 0$; H_1 : not all $\gamma_k = 0$

$F^* = MSC/MSE \sim F(c-1, abc(r-1))$ if H_0 is true

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Random Model

Source	E(MS) Random	F*
A	$\sigma^2 + r\sigma_{\alpha}^2 + cr\sigma_{\beta}^2 + br\sigma_{\gamma}^2 + bcr\sigma_{\alpha}^2$	
B	$\sigma^2 + r\sigma_{\alpha}^2 + cr\sigma_{\beta}^2 + ar\sigma_{\gamma}^2 + acr\sigma_{\beta}^2$	
C	$\sigma^2 + r\sigma_{\alpha}^2 + br\sigma_{\gamma}^2 + ar\sigma_{\beta}^2 + abr\sigma_{\gamma}^2$	
AB	$\sigma^2 + r\sigma_{\alpha}^2 + cr\sigma_{\beta}^2$	MSAB/MSABC
AC	$\sigma^2 + r\sigma_{\alpha}^2 + br\sigma_{\gamma}^2$	MSAC/MSABC
BC	$\sigma^2 + r\sigma_{\alpha}^2 + ar\sigma_{\beta}^2$	MSBC/MSABC
ABC	$\sigma^2 + r\sigma_{\alpha}^2$	MSABC/MSE
Error	σ^2	



How to construct the statistic to test each effect

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H_0 : all $\alpha\beta_{ij} = 0$; H_1 : not all $\alpha\beta_{ij} = 0$

$F^* = MSAB/MSE \sim F((a-1)(b-1), abc(r-1))$ if H_0 is true

H_0 : all $\alpha\gamma_{ik} = 0$; H_1 : not all $\alpha\gamma_{ik} = 0$

$F^* = MSAC/MSE \sim F((a-1)(c-1), abc(r-1))$ if H_0 is true

H_0 : all $\beta\gamma_{jk} = 0$; H_1 : not all $\beta\gamma_{jk} = 0$

$F^* = MSBC/MSE \sim F((b-1)(c-1), abc(r-1))$ if H_0 is true

H_0 : all $\alpha\beta\gamma_{ijk} = 0$; H_1 : not all $\alpha\beta\gamma_{ijk} = 0$

$F^* = MSABC/MSE \sim F((a-1)(b-1)(c-1), abc(r-1))$ if H_0 is true

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$H_0: \sigma_{\alpha}^2 = 0$; $H_1: \sigma_{\alpha}^2 > 0$

$F^* = MSA/MS_{C1} \sim F(a-1, df_{C1})$ if H_0 is true

$$MS_{C1} = MSAB + MSAC - MSABC$$

$$df_{C1} = \frac{(MSAB + MSAC - MSABC)^2}{\frac{(MSAB)^2}{df_{AB}} + \frac{(MSAC)^2}{df_{AC}} + \frac{(MSABC)^2}{df_{ABC}}}$$

$H_0: \sigma_{\beta}^2 = 0$; $H_1: \sigma_{\beta}^2 > 0$

$F^* = MSB/MS_{C2} \sim F(b-1, df_{C2})$ if H_0 is true

$$MS_{C2} = MSAB + MSBC - MSABC$$

$$df_{C2} = \frac{(MSAB + MSBC - MSABC)^2}{\frac{(MSAB)^2}{df_{AB}} + \frac{(MSBC)^2}{df_{BC}} + \frac{(MSABC)^2}{df_{ABC}}}$$

$H_0: \sigma_{\gamma}^2 = 0$; $H_1: \sigma_{\gamma}^2 > 0$

$F^* = MSC/MS_{C3} \sim F(c-1, df_{C3})$ if H_0 is true

$$MS_{C3} = MSAC + MSBC - MSABC$$

$$df_{C3} = \frac{(MSAC + MSBC - MSABC)^2}{\frac{(MSAC)^2}{df_{AC}} + \frac{(MSBC)^2}{df_{BC}} + \frac{(MSABC)^2}{df_{ABC}}}$$

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$H_0: \sigma_{\alpha\beta}^2 = 0$; $H_1: \sigma_{\alpha\beta}^2 > 0$

$F^* = MSAB/MS_{ABC} \sim F((a-1)(b-1), (a-1)(b-1)(c-1))$ if H_0 is true

$H_0: \sigma_{\alpha\gamma}^2 = 0$; $H_1: \sigma_{\alpha\gamma}^2 > 0$

$F^* = MSAC/MS_{ABC} \sim F((a-1)(c-1), (a-1)(b-1)(c-1))$ if H_0 is true

$H_0: \sigma_{\beta\gamma}^2 = 0$; $H_1: \sigma_{\beta\gamma}^2 > 0$

$F^* = MSBC/MS_{ABC} \sim F((b-1)(c-1), (a-1)(b-1)(c-1))$ if H_0 is true

$H_0: \sigma_{\alpha\beta\gamma}^2 = 0$; $H_1: \sigma_{\alpha\beta\gamma}^2 > 0$

$F^* = MSABC/MSE \sim F((a-1)(b-1)(c-1), abc(r-1))$ if H_0 is true

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Case Study (3)

- A study of the effects of sex, body fat and smoking history on exercise tolerance in stress testing of persons 25 to 35 years old.
- Factor:
 - sex (male, female)
 - Body Fat (low, high)
 - Smoking history (light, heavy)
 - 8 treatments (2×2×2)

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Data Test;
Input Sex \$ Fat \$ Smoking \$ Score@@;
Datalines;
A1 B1 C1 24 A1 B1 C2 18 A1 B1 C2 29 A1 B1 C2 19
A1 B1 C1 25 A1 B1 C2 23 A2 B1 C1 20 A2 B1 C2 15
A2 B1 C1 22 A2 B1 C2 10 A2 B1 C1 18 A2 B1 C2 11
A1 B2 C1 15 A1 B2 C2 15 A1 B2 C1 15 A1 B2 C2 20
A1 B2 C1 12 A1 B2 C2 13 A2 B2 C1 16 A2 B2 C2 10
A2 B2 C1 9 A2 B2 C2 14 A2 B2 C1 11 A2 B2 C2 6

PROC GLM;
Class Sex Fat Smoking;
MODEL Score= Sex|Fat|Smoking; Fixed Model
Means Sex /Tukey;
Means Fat /Tukey;
Means Smoking /Tukey;
LSMeans Sex*Fat / STDERR PDIF;
LSMeans Sex*Smoking /STDERR PDIF;
LSMeans Fat*Smoking /STDERR PDIF;
LSMeans Sex*Fat*Smoking /STDERR PDIF;
RUN;

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Source	DF	Squares	Mean Square	F Value	Pr > F
Model	7	610.5000000	87.2142857	9.97	<.0001
Sex	1	181.5000000	181.5000000	20.74	0.0003
Fat	1	253.5000000	253.5000000	28.97	<.0001
Smoking	1	73.5000000	73.5000000	8.40	0.0105
Sex*Fat	1	13.5000000	13.5000000	1.54	0.2321
Sex*Smoking	1	13.5000000	13.5000000	1.54	0.2321
Fat*Smoking	1	73.5000000	73.5000000	8.40	0.0105
Sex*Fat*Smoking	1	1.5000000	1.5000000	0.17	0.6843
Error	16	140.0000000	8.7500000		

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Tukey Grouping	Mean	N	Sex
A	19.000	12	A1
B	13.500	12	A2

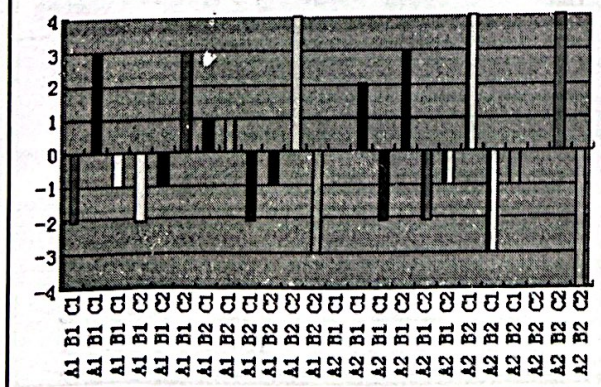
Fat	Smoking	Score LSMEAN	SE	Pr > t	Number
B1	C1	23.0000000	1.2076147	<.0001	1
B1	C2	16.0000000	1.2076147	<.0001	2
B2	C1	13.0000000	1.2076147	<.0001	3
B2	C2	13.0000000	1.2076147	<.0001	4

I\J	1	2	3	4
1		0.0008	<.0001	<.0001
2	0.0008		0.0981	0.0981
3	<.0001	0.0981		1.0000
4	<.0001	0.0981	1.0000	

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(±3) 个百. 其它可以去掉再算一次.

Residual Error



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Mixed model (correct EMS)

	Source	E(MS) Mixed
Sex	A	$\sigma_e^2 + r\sigma_{a\beta\gamma}^2 + c\sigma_{a\beta}^2 + b\sigma_{a\gamma}^2 + \frac{bcr}{a-1}\sum_1 a_i^2$
Fat	B	$\sigma_e^2 + a\sigma_{\beta\gamma}^2 + ac\sigma_{\beta}^2$
Smoke	C	$\sigma_e^2 + a\sigma_{\beta\gamma}^2 + ab\sigma_{\gamma}^2$
	AB	$\sigma_e^2 + r\sigma_{a\beta\gamma}^2 + c\sigma_{a\beta}^2$
	AC	$\sigma_e^2 + r\sigma_{a\beta\gamma}^2 + b\sigma_{a\gamma}^2$
	BC	$\sigma_e^2 + a\sigma_{\beta\gamma}^2$
	ABC	$\sigma_e^2 + r\sigma_{a\beta\gamma}^2$
	Error	σ_e^2

上述实例: $a=b=c=2, r=3$

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SAS Gives Wrong EMS

Source	Type III Expected Mean Square
Sex	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(S^*M)+6\text{Var}(S^*F)+Q(S)$
Fat	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(F^*M)+6\text{Var}(S^*F)+12\text{Var}(F)$
Smoke	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(F^*M)+6\text{Var}(S^*F)+12\text{Var}(M)$
S*F	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(S^*F)$
S*M	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(S^*M)$
F*M	$\text{Var}(E)+3\text{Var}(S^*F^*M)+6\text{Var}(F^*M)$
S*F*M	$\text{Var}(E)+3\text{Var}(S^*F^*M)$

Note: S, F, M 分别指 Sex, Fat, Smoke

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SAS program (Mixed model)

```
PROC GLM;
Class Sex Fat Smoking;
MODEL Score= Sex|Fat|Smoking/P;
Random Fat Smoking Sex*Fat
Sex*Smoking Fat*Smoking
Sex*Fat*Smoking / test;
Test H = Fat Smoking E = Fat*Smoking;
Means Sex /Tukey;
RUN;
```

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Tests of Hypotheses for Mixed Model Analysis of Variance

Source	DF	Squares	Mean Square	F Value	Pr > F
Sex	1	181.500000	181.500000	7.12	0.1319
Error	1.773	45.211656	25.500000		

Error: $MS(S^*F)+MS(S^*M)-MS(S^*F^*M)+11E-17*MS(E)$

Fat	1	253.500000	253.500000	2.96	0.2882
Error	1.3085	111.876561	85.500000		

Error: $MS(S^*F)+MS(F^*M)-MS(S^*F^*M)+11E-17*MS(E)$

Smoke	1	73.500000	73.500000	0.86	0.4927
Error	1.3085	111.876561	85.500000		

Error: $MS(S^*F)+MS(F^*M)-MS(S^*F^*M)+11E-17*MS(E)$

Sex*Fat	1	13.500000	13.500000	9.00	0.2048
Sex*Smoke	1	13.500000	13.500000	9.00	0.2048
Fat*Smoke	1	73.500000	73.500000	49.00	0.0903
Error	1	1.500000	1.500000		

Error: $MS(S^*F^*M)$

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Estimates of variances

Cov Parm	Estimate
Fat	14.7665
Smoking	0
Sex*Fat	0.8624
Sex*Smoking	0.8483
Fat*Smoking	10.4670
Sex*Fat*Smoking	0
Residual	8.3255

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Case Study (4)

- ☐ Cotton Testing for variety, planting time and stands:
- ☐ Factors (Mixed):
 - Random Block (3 Replicates)
 - Qualitative A: Varieties (A1 A2)
 - Qualitative B: Times (B1 B2)
 - Quantitative Stands (3500P/Mu, 5000P/Mu, 6500P/Mu)
 - 12 treatments ($2 \times 2 \times 3$)
- ☐ Experimental units: 36 plots (25m²) in 3 blocks
- ☐ Response: Yield (kg)

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表 13.16 棉花三因素试验的各处理

A 品种	B 播期	密度	处理组合代号
A ₁	B ₁ (谷雨前)	C ₁ (3500)	T ₁
		C ₂ (5000)	T ₂
		C ₃ (6500)	T ₃
	B ₂ (立夏前)	C ₁ (3500)	T ₄
		C ₂ (5000)	T ₅
		C ₃ (6500)	T ₆
A ₂	B ₁ (谷雨前)	C ₁ (3500)	T ₇
		C ₂ (5000)	T ₈
		C ₃ (6500)	T ₉
	B ₂ (立夏前)	C ₁ (3500)	T ₁₀
		C ₂ (5000)	T ₁₁
		C ₃ (6500)	T ₁₂

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图 13.2 棉花三因素随机区组试验的田间排列和产量 (kg/25m²)

区组 I	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	T ₈	T ₉	T ₁₀	T ₁₁	T ₁₂
1	12	9	7	5	10	4	12	2	10	3	3	6
区组 II	T ₁₂	T ₁₁	T ₁₀	T ₉	T ₈	T ₇	T ₆	T ₅	T ₄	T ₃	T ₂	T ₁
2	7	2	11	14	14	16	6	2	3	9	9	9
区组 III	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆	T ₇	T ₈	T ₉	T ₁₀	T ₁₁	T ₁₂
3	9	13	5	11	7	7	8	3	7	4	4	9

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表 13.17 区组和处理双向表

A 品种	B 肥料制	密 度	区 组			T _总
			I	II	III	
A ₁	B ₁	C ₁	12	14	15	39
		C ₂	12	11	11	34
		C ₃	10	9	9	28
	B ₂	C ₁	10	9	9	28
		C ₂	9	9	8	26
		C ₃	6	6	7	19
	B ₃	C ₁	3	3	4	9
		C ₂	4	3	4	11
		C ₃	7	6	7	20
B ₄	C ₁	2	2	3	7	
	C ₂	3	4	5	12	
	C ₃	5	7	7	19	
T _总			83	82	87	T=252

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表 13.18 二因素双向表

(1) AB 双向表				(2) AC 双向表				(3) BC 双向表			
	B ₁	B ₂	T ₁		C ₁	C ₂	T ₁		C ₁	C ₂	T ₁
A ₁	100	73	174	A ₁	67	60	127	B ₁	48	45	93
A ₂	40	38	78	A ₂	16	23	39	B ₂	35	38	73
T ₁	140	111	T=252	T ₁	83	83	T=166	T ₁	83	83	T=166

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Data Cotton;
 Input Variety \$ Time \$ Stand \$ Block \$
 Yield @@;
 Datalines;
 A1 B1 C1 R1 12 A1 B1 C1 R2 14 A1 B1 C1 R3 13
 A1 B1 C2 R1 12 A1 B1 C2 R2 11 A1 B1 C2 R3 11
 A1 B1 C3 R1 10 A1 B1 C3 R2 9 A1 B1 C3 R3 9
 A1 B2 C1 R1 10 A1 B2 C1 R2 9 A1 B2 C1 R3 9
 A1 B2 C2 R1 9 A1 B2 C2 R2 9 A1 B2 C2 R3 8
 A1 B2 C3 R1 6 A1 B2 C3 R2 6 A1 B2 C3 R3 7
 A2 B1 C1 R1 3 A2 B1 C1 R2 2 A2 B1 C1 R3 4
 A2 B1 C2 R1 4 A2 B1 C2 R2 3 A2 B1 C2 R3 4
 A2 B1 C3 R1 7 A2 B1 C3 R2 6 A2 B1 C3 R3 7
 A2 B2 C1 R1 2 A2 B2 C1 R2 2 A2 B2 C1 R3 3
 A2 B2 C2 R1 3 A2 B2 C2 R2 4 A2 B2 C2 R3 5
 A2 B2 C3 R1 5 A2 B2 C3 R2 7 A2 B2 C3 R3 7
 ;

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```

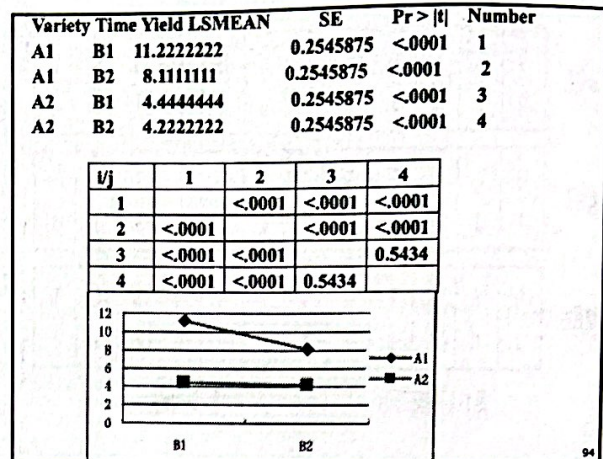
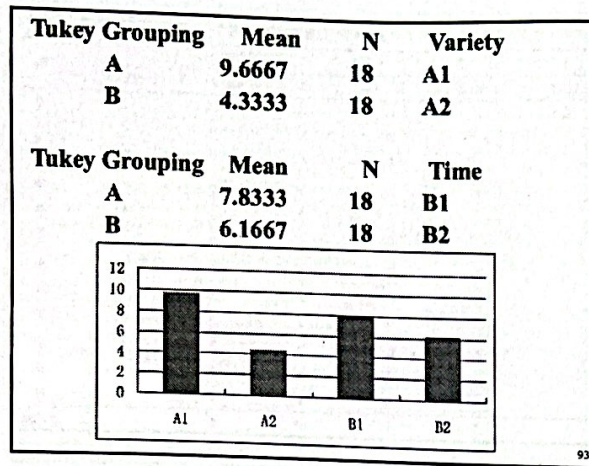
PROC GLM;
Class Variety Time Stand Block;
MODEL Yield = Variety|Time|Stand Block;
Contrast 'Linear' Stand -1 0 1;
Contrast 'Quadratic' Stand 1 -2 1;
Means Variety /Tukey;
Means Time /Tukey;
LSMeans Variety*Time / STDERR PDIF;
LSMeans Variety*Stand / STDERR PDIF;
LSMeans Time*Stand / STDERR PDIF;
LSMeans Variety*Time*Stand / STDERR PDIF;
RUN;
    
```

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Result of PROC GLM

Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Model	13	383.1666667	29.4743590	50.53	<.0001
Variety	1	256.0000000	256.0000000	438.86	<.0001
Time	1	25.0000000	25.0000000	42.86	<.0001
Stand	2	0.5000000	0.2500000	0.43	0.6568
Variety*Time	1	18.7777778	18.7777778	32.19	<.0001
Variety*Stand	2	80.1666667	40.0833333	68.71	<.0001
Time*Stand	2	1.5000000	0.7500000	1.29	0.2964
Variety*Time*Stand	2	0.0555556	0.0277778	0.05	0.9536
Block	2	1.1666667	0.5833333	1.00	0.3840
Error	22	12.8333333	0.5833333		

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Regression Approach

Var	Time	Std_L	Std_Q	Yld	Var	Time	Std_L	Std_Q	Yld	Var	Time	Std_L	Std_Q	Yld
1	-1	-0.707107	0.408248	12	1	-1	-0.707107	0.408248	14	1	-1	-0.707107	0.408248	13
1	-1	0	-0.816497	12	1	-1	0	-0.816497	11	1	-1	0	-0.816497	11
1	-1	0.707107	0.408248	10	1	-1	0.707107	0.408248	9	1	-1	0.707107	0.408248	9
1	1	-0.707107	0.408248	10	1	1	-0.707107	0.408248	9	1	1	-0.707107	0.408248	9
1	1	0	-0.816497	9	1	1	0	-0.816497	9	1	1	0	-0.816497	8
1	1	0.707107	0.408248	6	1	1	0.707107	0.408248	6	1	1	0.707107	0.408248	7
-1	-1	-0.707107	0.408248	3	-1	-1	-0.707107	0.408248	2	-1	-1	-0.707107	0.408248	4
-1	-1	0	-0.816497	4	-1	-1	0	-0.816497	3	-1	-1	0	-0.816497	4
-1	-1	0.707107	0.408248	7	-1	-1	0.707107	0.408248	6	-1	-1	0.707107	0.408248	7
-1	1	-0.707107	0.408248	2	-1	1	-0.707107	0.408248	2	-1	1	-0.707107	0.408248	3
-1	1	0	-0.816497	3	-1	1	0	-0.816497	4	-1	1	0	-0.816497	5
-1	1	0.707107	0.408248	5	-1	1	0.707107	0.408248	7	-1	1	0.707107	0.408248	7

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Data Cotton;
Input Variety Time S_L S_Q Yield @@;
Datalines;

```

1 -1 -0.707107 0.408248 12 1 -1 -0.707107 0.408248 14 1 -1 -0.707107 0.408248 13
1 -1 0 -0.816497 12 1 -1 0 -0.816497 11 1 -1 0 -0.816497 11
1 -1 0.707107 0.408248 10 1 -1 0.707107 0.408248 9 1 -1 0.707107 0.408248 9
1 1 -0.707107 0.408248 10 1 1 -0.707107 0.408248 9 1 1 -0.707107 0.408248 9
1 1 0 -0.816497 9 1 1 0 -0.816497 9 1 1 0 -0.816497 8
1 1 0.707107 0.408248 6 1 1 0.707107 0.408248 6 1 1 0.707107 0.408248 7
-1 -1 -0.707107 0.408248 3 -1 -1 -0.707107 0.408248 2 -1 -1 -0.707107 0.408248 4
-1 -1 0 -0.816497 4 -1 -1 0 -0.816497 3 -1 -1 0 -0.816497 4
-1 -1 0.707107 0.408248 7 -1 -1 0.707107 0.408248 6 -1 -1 0.707107 0.408248 7
-1 1 -0.707107 0.408248 2 -1 1 -0.707107 0.408248 2 -1 1 -0.707107 0.408248 3
-1 1 0 -0.816497 3 -1 1 0 -0.816497 4 -1 1 0 -0.816497 5
-1 1 0.707107 0.408248 5 -1 1 0.707107 0.408248 7 -1 1 0.707107 0.408248 7

```

前一程序A1, A2 用 1. -1
 B1, B2 用 -1. 1
 C1 用 -0.707107 0.408248
 C2 用 0 -0.816497
 C3 用 0.707107 0.408248

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```

PROC GLM;
MODEL Yield = Variety Time
Variety*Time
S_L S_Q Variety*S_L Variety*S_Q Time*S_L
Time*S_Q;
RUN;

PROC GLM;
MODEL Yield = Variety Time Variety*Time
Variety*S_L;
RUN;

```

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Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Variety	1	256.0000000	256.0000000	473.55	<.0001
Time	1	25.0000000	25.0000000	46.25	<.0001
S_L	1	0.3750000	0.3750000	0.69	0.4125
S_Q	1	0.1250000	0.1250000	0.23	0.6346
Variety*Time	1	18.7777778	18.7777778	34.74	<.0001
Variety*S_L	1	77.0416667	77.0416667	142.51	<.0001
Variety*S_Q	1	3.1250000	3.1250000	5.78	0.0236
Time*S_L	1	0.3750000	0.3750000	0.69	0.4125
Time*S_Q	1	1.1250000	1.1250000	2.08	0.1611
Error	26	14.0555556	0.5405983		

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Parameter	Estimate	Error	tValue	Pr> t
Intercept	7.000000034	0.12254232	57.12	<.0001
Variety	2.666666497	0.12254232	21.76	<.0001
Time	-0.833333435	0.12254232	-6.80	<.0001
S_L	0.176776641	0.21224945	0.83	0.4125
S_Q	0.102062062	0.21224949	0.48	0.6346
Variety*Time	-0.722222222	0.12254232	-5.89	<.0001
Variety*S_L	-2.533798515	0.21224945	-11.94	<.0001
Variety*S_Q	-0.510310309	0.21224949	-2.40	0.0236
Time*S_L	0.176776641	0.21224945	0.83	0.4125
Time*S_Q	-0.306186186	0.21224949	-1.44	0.1611

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Source	DF	Sum Squares	Mean Square	F Value	Pr > F
Variety	1	256.0000000	256.0000000	413.75	<.0001
Time	1	25.0000000	25.0000000	40.41	<.0001
Variety*Time	1	18.7777778	18.7777778	30.35	<.0001
Variety*S_L	1	77.0416667	77.0416667	124.52	<.0001
Variety*S_Q	1	3.1250000	3.1250000	5.84	0.0220

Parameter	Estimate	Error	T Value	Pr> t
Intercept	7.000000000	0.13109873	53.39	<.0001
Variety	2.666666667	0.13109873	20.34	<.0001
Time	-0.833333333	0.13109873	-6.36	<.0001
Variety*Time	-0.722222222	0.13109873	-5.51	<.0001
Variety*S_L	-2.533798515	0.22706960	-11.16	<.0001
Variety*S_Q	-0.510310309	0.21118418	-2.42	0.0220

$\hat{Y} = 7.00 + 2.667V - 0.833T - 0.7222V \times T$
 $- 2.534V \times S_L - 0.510V \times S_Q$

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□ C factor has three levels: 3500, 5000, 6500.

$m = 5000, \Delta = 1500, k = 3, \lambda_1 = 1, C_1 = 2, \lambda_2 = 3, C_2 = 6$

□ At the stand of S (3500~6500)

$$S_L = \left(\frac{S - 5000}{1500} \right) / \sqrt{2};$$

$$S_Q = 3 \left[\left(\frac{S - 5000}{1500} \right)^2 - \left(\frac{3^2 - 1}{12} \right) \right] / \sqrt{6}$$

Level	P_1	P_2
1	-1	1
2	1	-1
3	0	0

□ Predicted Yield

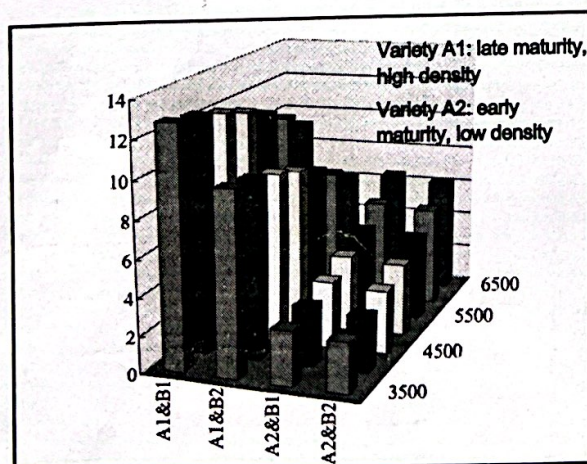
$$\hat{Y} = 7.00 + 2.667V - 0.833T - 0.7222V \times T$$

$$- 2.534V \times S_L - 0.510V \times S_Q$$

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A	B	C	S_L	S_Q	Y	A	B	C	S_L	S_Q	Y
1	-1	3500	-0.707	0.408	12.806	-1	-1	3500	-0.707	0.408	2.86
1	-1	4000	-0.471	-0.272	12.555	-1	-1	4000	-0.471	-0.272	3.111
1	-1	4500	-0.236	-0.68	12.166	-1	-1	4500	-0.236	-0.68	3.5
1	-1	5000	0	-0.816	11.638	-1	-1	5000	0	-0.816	4.028
1	-1	5500	0.236	-0.68	10.972	-1	-1	5500	0.236	-0.68	4.694
1	-1	6000	0.471	-0.272	10.166	-1	-1	6000	0.471	-0.272	5.5
1	-1	6500	0.707	0.408	9.222	-1	-1	6500	0.707	0.408	6.444
1	1	3500	-0.707	0.408	9.696	-1	1	3500	-0.707	0.408	2.638
1	1	4000	-0.471	-0.272	9.445	-1	1	4000	-0.471	-0.272	2.889
1	1	4500	-0.236	-0.68	9.056	-1	1	4500	-0.236	-0.68	3.278
1	1	5000	0	-0.816	8.528	-1	1	5000	0	-0.816	3.806
1	1	5500	0.236	-0.68	7.862	-1	1	5500	0.236	-0.68	4.472
1	1	6000	0.471	-0.272	7.056	-1	1	6000	0.471	-0.272	5.278
1	1	6500	0.707	0.408	6.112	-1	1	6500	0.707	0.408	6.222

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